

Advances in dynamic energy budget modeling of terrestrial arthropods by considering different timing of reproductive investment

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ABSTRACT

Linking Dynamic Energy Budget (DEB) models with toxicokinetic-toxicodynamic (TKTD) modules allows for an integrated mechanistic interpretation of lethal and sublethal effects of a species exposed to a toxicant, which has been proposed as a refinement option in the chronic risk assessment of pesticides. Yet, terrestrial non-target arthropods (NTAs) are an underrepresented group when it comes to readily available DEB models in the Addmy-Pet (AmP) portal. One reason for this is that many insect species have complex life cycles, particularly with regard to the acquisition and turnover of reproductive energy, that cannot be adequately represented by existing predefined DEB model variants.

In this study we attempted to parametrize DEB models for four regulatory relevant NTAs, the predatory mite *Typhlodromus pyri*, the parasitic wasp *Aphidius rhopalosiphii*, the seven-spot ladybird beetle *Coccinella septempunctata*, and the green lacewing *Chrysoperla carnea* with data obtained from open literature and additional laboratory experiments. While parametrization of *T. pyri* was not possible due to a lack of essential calibration data, the other species could be successfully parametrized through appropriate model adaptations. The life cycle of *A. rhopalosiphii* could be accurately modeled with the hax model variant, a predefined DEB model for holometabolous insects, which assumes that energetic investment in reproduction mainly takes place during the larval phase. A parasitization module was added to this model allowing to simulate the parasitization rate of *A. rhopalosiphii* depending on several known influencing factors: the initial reproduction buffer size, host density, host type, temperature, and timing during the reproductive phase. For both *C. septempunctata* and *C. carnea*, a new model variant was developed which, in contrast to the existing models for holometabolous insects, assumes that the energetic investment in the formation of eggs takes place mainly after the emergence of the imago. We propose this new model variant as a template for other terrestrial insect species with life-history strategies similar to *C. septempunctata* and *C. carnea*. Due to the good representation of the calibration data and a well struck balance between model accuracy and model complexity, we consider all three models suitable for use in DEB-TKTD modeling for regulatory purposes.

1. Introduction

Bioenergetic models based on the Dynamic Energy Budget (DEB) theory are used today in various domains ranging from basic research to applications in environmental conservation and economic applications, e.g., in the field of aquaculture (Haberle et al., 2024; Lavaud et al., 2021; Yang et al., 2020). The core concept of DEB models is that they can predict the development, growth, feeding, reproduction, and aging of

organisms over time depending on variable external factors such as temperature and food supply, while strictly adhering to basic physical principles such as the laws of conservation of energy and mass (Kooijman, 2010a). This makes it possible, for example, to predict complex population dynamics under temporally heterogeneous environmental conditions (Accolla and Forbes, 2021). Due to their generality, DEB models are applicable to a wide range of species including insects (Llandres et al., 2015; Gergs and Baden, 2021; Klagkou et al.,

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2024; Koch et al., 2024).

One important application of DEB models is their use in ecotoxicology and the environmental risk assessment of chemicals, which is also the field for which the DEB theory had been originally conceptualized (Kooijman and Metz, 1984). Here, DEB models can be linked with toxicokinetic-toxicodynamic (TKTD) models (DEB-TKTD) to make predictions about lethal and sublethal effects over time for a given species-substance combination (Matyja et al., 2020; Jager et al., 2023; Romoli et al., 2024). This allows acute and chronic effects to be predicted efficiently under various environmental conditions, which otherwise would be difficult and costly to test experimentally. Simulations at time-variable conditions of temperature, food availability, and chemical exposure, as they occur in the field, are also easily possible (Sherborne et al., 2020; Koch et al., 2024). In pesticide risk assessment, EFSA has identified DEB-TKTD models as scientifically appropriate tools for refining chronic effect assessments in aquatic organisms (EFSA, 2018) and in birds and mammals (EFSA, 2023), while noting that their use remains subject to expert judgment, and more case-studies and user-friendly tools are needed before routine regulatory application.

In order to calibrate a DEB-TKTD model to reproduce experimentally observed toxic effects adequately, at first the normal physiological behavior of a species at unstressed control conditions must be explained well enough by the DEB part of the model (Jager et al., 2023). The DEB model simulates the entire life history of the organism based on clearly defined rules for the energy uptake from food and its allocation to key physiological processes (growth, maintenance, maturation, reproduction). While species-specific physiological rates are represented by different parametrizations, different life cycles are covered by using different model variants (Marques et al., 2018). In addition to the above mentioned need for an accurate control model to serve as a reference when conducting a DEB-TKTD parametrization, selecting an appropriate model structure for a species' life cycle is also critical when extrapolating from individuals to populations using population models. For insect populations, for example, Gurney & Nisbet (1985) demonstrated how different model assumptions regarding the effects of larval food competition on larval and pupal survival, development time, and adult fecundity led to distinct types of population cycles, underscoring the importance of model detail at the individual level.

The Add-my-Pet (AmP) portal (hyperlink: https://www.bio.vu.nl/thb/deb/deblab/add_my_pet/) offers a multitude of fully parametrized DEB models for individual species, including many species that are relevant in a regulatory context for pesticide registration. This is due in large part to the efforts of the AmP community, which aims to lower the hurdle for the parametrization of new DEB models by interested users as much as possible. In addition to regular training courses and the provision of free software tools, extensive literature on clearly predefined DEB model variants helps users to select the model type that best suits their own species (Kooijman, 2010b). A major advantage of the predefined models is that they allow characteristic life cycles of different species to be grouped together (Marn and Kooijman, 2022; Llandres et al., 2015; Kooijman, 2014). This simplifies model documentation, the comparability of different species with similar life cycles at parameter level, and facilitates the comprehensibility of the models.

Notably, lots of regulatory relevant non-target arthropods (NTAs) are still missing from the AmP database. A likely reason for this is that many arthropod species have complex life-history strategies that cannot be adequately represented by the predefined DEB model structures provided in the DEB literature at the time of writing (Kooijman, 2010b). As an example, all DEB model variants for holometabolous insects established so far (i.e., the model types *hex*, and *hax*) assume that the organisms acquire their entire reproductive energy during the larval stage. Therefore, species that invest energy in reproduction partially or exclusively after the imago has emerged are currently not covered by any standardized DEB model variant.

Since the energetic investment in reproduction cannot be measured directly, we use *ovigeny*, a concept primarily used to characterize egg

maturation patterns in parasitic wasps (Jervis et al., 2001) as an indicator of when the main investment in reproduction occurs in a holometabolous insect species. The fraction of fully-developed eggs at adult emergence relative to the potential lifetime fecundity is termed the *ovigeny index* (OI). Whereas species with an OI = 1 are termed *pro-ovigenic*, all species with an OI < 1 are termed *synovigenic*. *Extremely synovigenic* species have an OI = 0, which means that no mature eggs are present at the time of adult emergence. Although for pro-ovigenic species and some synovigenic species with an OI close to 1, the current default DEB model assumption that all reproductive investment takes place during the larval phase may be valid, new model variants are needed to adequately represent synovigenic species with a lower OI, particularly those with an OI of zero.

The aim of this study was to parametrize DEB models for four regulatory-relevant NTAs: the predatory mite *Typhlodromus pyri*, the parasitic wasp *Aphidius rhopalosiphii*, the seven-spot ladybird *Coccinella septempunctata*, and the green lacewing *Chrysoperla carnea* with data obtained from open literature and additional laboratory experiments. During parametrization, preference was given to the already established DEB model variants, but whenever the life cycles of species could not be adequately represented by the predefined models, especially when species were identified to have life history strategies like synovigeny, emphasis was placed in this study on capturing these through appropriate model extensions or adaptations.

2. Material and methods

2.1. Literature search

An exhaustive literature search for all four species of interest was conducted following the guidance provided by EFSA (2011). The search was targeted at life-history data that can be used for the parametrization of DEB models, including data on developmental times, growth, reproduction, metabolic rate, and longevity, ideally at different temperatures and feeding regimes. For all species, the data search focused on data on females. For life stages during which sexual differentiation was not yet possible, data of undifferentiated sex were considered. A detailed documentation of the search parameters, including used search engines, search terms, and search dates can be found in the supplementary material.

In addition to this online literature search, we made use of literature on *C. septempunctata* that was available to us as hard copy, control parasitization numbers from Bayer in-house standard tests with *A. rhopalosiphii*, and new data from growth experiments with *C. septempunctata* and *Chrysoperla carnea* (see next section).

2.2. Additional growth experiments

The body weight, length and width of *C. septempunctata* and *C. carnea* were measured in two test runs each under controlled laboratory conditions throughout their development. Both organisms were sourced from an external breeder (Katz Biotech AG). As test unit a steel ring (diameter 4.1 cm, height 3 cm, top of inner edge coated with lubricant (PTFE)) was placed upon a leaf disc, which was laid on a wet cotton wool pad in a petri dish (diameter 9 cm). Freshly hatched larvae (less than 24 h) were haltered individually. Sterilized eggs of *Ephesttia kuehniella* (for *C. carnea*) and mixed population of *Acyrtosiphon pisum* (for *C. septempunctata*) were provided as food *ad libitum*. During the entire experiment the mean temperature was 25 °C (± 2 °C), mean relative humidity was 60–90 % with a light:dark cycle of 16:8 h LD. For the first 7 (*C. carnea* 2nd run) or 8 (*C. carnea* 1st run, *C. septempunctata*) days of the experiments, a sample of 10 larvae was randomly taken. Pictures of larvae in an empty petri dish placed on graph paper were taken to determine the individual body length and width using GIMP (2.10.36) (The GIMP Development Team, 2024). All 10 larvae were pooled for weight determination. The same assessment was performed during the

pupal stage, but only on the first day after pupation for *C. septempunctata* and on days 9, 14, 17 (1st run) or days 8, 11, 15 (2nd run) for *C. carnea*. Weight (pooled), length, and width (individual) of 10 adults of *C. carnea* were measured on day 18 (1st run) or 17 (2nd run), respectively. On days 14 and 21 or 24 (*C. septempunctata*), 22 and 31 (1st run *C. carnea*), or 26 and 32 (2nd run *C. carnea*), weight, length and width of 10 males and females each were measured individually, if possible. Individuals were only subject of measurement once and discharged afterwards.

2.3. DEB model variants and adaptations

Generally, typical DEB model variants exist for different species groups that often lead to appropriate results. While to our knowledge there is no rigorous decision tree on how to arrive at a suitable model choice, the AmP database can help to identify plausible model variants for the species under investigation by searching for related species. If these models do not lead to satisfactory results, closely related model variants can be explored or new model variants need to be developed *ad hoc*.

For *T. pyri*, the data availability was considered insufficient after the literature search (see Results and Discussion), which is why no model parametrization was carried out for the time being. All three other species in this study are holometabolous insects, which is why the two model variants hex and hax, defined specifically for holometabolous insects, were considered first. However, both model variants assume that all energy that will be invested in the production of eggs is collected during the larval phase of the species (Kooijman, 2010b), an assumption that does not hold true for all insect species. Whereas pro-ovigenic species, i.e., those with an $OI = 1$, are conceptually represented well by the model types hex and hax, synovigenic species, i.e., those that

continue to mature oocytes during the adult stage ($OI < 1$), may require some adaptation to the existing model types. One source of uncertainty here is that the quantity of matured eggs is not directly representative of the energetic buffer *intended* for production of eggs. For species with an OI relatively close to 1, the default assumption of the model types hex and hax may still be valid, assuming that while not all eggs are fully developed at adult emergence, the energy intended for production of eggs was still fully determined during the larval phase. For other species with a low ovigeny index or with a high capability to adapt the egg maturation strategy to changing environmental conditions, more complex new model types may be needed, i.e., such that allow for some reproductive investment during the larval phase, as well as during the imago stage. However, such models would require highly detailed data to be correctly parametrized (i.e., ideally data on reproductive investment during the larval phase, data on the reproductive investment during the adult phase, and a clear understanding of how both contribute to the overall oviposition rate). Therefore, we deem it most appropriate to select a simplifying model type depending on whether reproduction investment *before* or *after* adult emergence makes the greatest contribution to the lifetime reproductive output, according to current evidence. In the first case, the model types hex or hax may be applied, while in the second case, a new model assuming reproductive investment only after adult emergence, like the new model variant presented in this study, is needed.

All three model variants considered in this study to describe the physiological life cycle of holometabolous insects are schematically represented in Fig. 1.

2.3.1. The hex and hax models for pro-ovigenic holometabolous insects

Both the hex and hax model variants have been described in detail in

hax variant	Egg	Larva		Pupa	Imago	
		Early	Late			
Feeding	No	Yes	Yes	No	Yes/No	
Growth	Normal	Accelerated	Normal	Normal	No	
Maturation	Yes	Yes	No	Yes	No	
Reproductive investment	No	No	Yes	No	No	

hex variant	Egg	Larva		Pupa	Imago	
Feeding	No	Yes		No	Yes/No	
Growth	Normal	Accelerated		Normal	No	
Maturation	Yes	No		Yes	No	
Reproductive investment	No	Yes		No	No	

New model variant	Egg	Larva		Pupa	Imago	
					Pre-oviposition	Oviposition
Feeding	No	Yes		No	Yes	Yes
Growth	Normal	Accelerated		Normal	No	No
Maturation	Yes	Yes		Yes	Yes	No
Reproductive investment	No	No		No	No	Yes

Fig. 1. Comparison scheme for the existing DEB model variants for holometabolous insects hax and hex, and the new model variant presented in this study. The scheme illustrates for each consecutive functional life stage whether or not the organism feeds, grows, matures, and invests energy in reproduction. The life stage during which investment in reproduction occurs in each respective model is highlighted in yellow. The pupa phase is highlighted in grey to clearly illustrate whether reproductive investment is assumed to occur before or after the pupa phase. *Normal* growth refers to the growth pattern of the standard DEB model, yielding a von Bertalanffy-type growth curve under constant food. *Accelerated* growth indicates a growth acceleration commonly observed in insect larvae. Imagines of the hax and hex model variants may or may not feed depending on the species.

the DEB theory literature (Llandres et al., 2015; Kooijman, 2010a; 2010b). A precise mathematical description of the hax model (which was ultimately selected as the best fit for *A. rhopalosiphi*) can be found in the supplementary material. In brief, in the hex model, the organism begins investing energy in the reproduction buffer as soon as it starts external feeding (*birth* in DEB terminology) while no additional investment in maturity is assumed during the larval stage. It feeds and grows with metabolic acceleration (Kooijman, 2014), resulting in an accelerated growth over time compared to the von Bertalanffy-type growth curve that emerges from the standard DEB model under constant food (see, e.g., Kearney, 2021). The larval growth phase ends at pupation, which is initiated when a predefined threshold of reproduction buffer per structural volume is reached. The hax model by contrast assumes that after its birth the organism invests energy in maturity first until a maturity threshold for *puberty* (DEB terminology) is reached. During this initial development phase, the organism also follows the metabolic acceleration growth pattern. At puberty, the organism starts investing energy in the reproduction buffer and the metabolic acceleration ends (i.e., the growth curve transitions to the von Bertalanffy pattern). Like in the hex model, pupation is triggered by a critical threshold of reproduction buffer per structural volume. After this, both model variants behave the same: at pupation, all body structure is transformed into reserve energy, and the maturity level is reset to zero. The reproduction buffer remains constant during the pupa phase. The pupa does not feed externally but draws from its reserve to regrow structure and mature. Adult emergence is again triggered by a threshold of maturity. The imago may or may not feed (this model decision should be based on the feeding behavior of the real organism).

2.3.2. A new parasitization submodel for *A. rhopalosiphi*

In both model variants hex and hax, all reproductive investment is assumed to occur during the larval phase. However, how egg formation and deposition occur during the adult phase as a function of time is not a formal part of the model structure. As a result of evolutionary differentiation, different strategies for oogenesis and egg deposition have evolved, that are highly species-specific, and depend on many physiological and external factors (see, e.g., Stamp 1980; Hughes and Simons, 2014; Johansson et al., 2023). Therefore, the host parasitization rate of *A. rhopalosiphi*, much like the egg deposition rates and patterns of non-parasitic insects, is a process that is not explicitly covered by the DEB theory. However, because the parasitization rate of the wasps in a fixed time window is one of the most frequently considered regulatory endpoints for *A. rhopalosiphi* (Mead-Briggs et al., 2000; 2010), we extended the core DEB model with a simple parasitization module, allowing us to estimate the parasitization rate as a function of multiple influencing factors. The module was based on the observations that (a) the parasitization rate has a linear phase, followed by a gradual decline phase (Shirota et al., 1983), (b) within the tolerance range, higher temperatures lead to higher parasitization rates during the linear phase (Sigsgaard, 2000), (c) the parasitization rate during the linear phase depends on host density following a Holling type III functional response (Shirota et al., 1983), (d) the parasitization rate during the linear phase depends on host type (Pungerl, 1984), as well as the implicit but untested assumptions that (a) lower parasitization rates lead to a longer duration of the parasitization phase if the initial reproduction buffer size is the same, (b) temperature and host density affect the parasitization rate during the decline phase in the same way (i.e., with the same factor) as during the linear phase, and (c) only the maximum parasitization rate but not the shape of the Holling type III relationship is temperature- and host species dependent. A precise mathematical description of this module can be found in the supplementary material.

2.3.3. A new model variant for (extremely) synovigenic holometabolous insects

While the structure of the new model variant combines elements from several of the predefined models, we define it here as an extension

of the model type *abp* (Kooijman, 2010b), since it behaves identically to this model until the end of the larval phase: In the *abp* model an organism experiences a phase of metabolic acceleration between birth and the end of the juvenile growth phase. During this time, the organism invests energy in maturity and does not invest in reproduction. The end of the growth phase is determined by a threshold of maturity. While the end of growth in the *abp* model normally also means the end of maturation and the start of reproduction, we extend the model here to include the pupal stage, which behaves identically to the hex and hax model types described above. After adult emergence, we assume a final maturation phase representing the preoviposition phase, during which the ovaries develop and no egg-laying takes place. In this phase, the emerged animal no longer grows, but feeds again and invests energy in maturity until a final maturity threshold is reached. This moment marks the beginning of the reproductive phase, during which the model again behaves like the *abp* model when sexual maturity is reached: the individual feeds, no longer grows, but invests all energy not needed for maintenance in the formation of eggs. Assuming that the egg-laying rate corresponds approximately to the continuous investment in the reproduction buffer, reproduction rates can be simulated in this model without the need for an additional module like the one for *A. rhopalosiphi* described in the previous section. A precise mathematical description of this new model variant can be found in the supplementary material.

2.3.4. Parameter estimation

The parameter estimation was done for each species separately using the software tool DEBtool in MATLAB (R2017b) (The MathWorks Inc. 2017), which can be downloaded for free from the AmP portal (hyperlink: https://www.bio.vu.nl/thb/deb/deblab/add_my_pet/). First, the code required to simulate the experimental data types with the respective model variant was written, also taking into account experimental conditions such as different test temperatures and feeding regimes. Whenever possible, the standard functions integrated in DEBtool were used. However, customized calculations were made for all non-standard model assumptions. Next, the parameter values were optimized so that the model simulations match the observed data as closely as possible. For this, a Nelder-Mead simplex routine embedded in DEBtool was employed. DEBtool's default loss function "multiplicative symmetric bounded" was used to guide this routine. Some parameters were set to typical default values. This is common practice in DEB parametrization when it is assumed that the respective parameter cannot be reliably estimated based on the available data. After successful calibration, all three models (including the calibration data, the optimized parameters, and the code for re-simulating the calibration data) were submitted to the AmP portal (hyperlink: https://www.bio.vu.nl/thb/deb/deblab/add_my_pet/) where they can be viewed and downloaded for free.

3. Results and discussion

3.1. Literature search

For each of the four species, a complete list of the literature identified as potentially relevant for model parametrization can be found in the supplementary material. In summary, many data types considered useful for DEB parametrization (see, e.g., Kooijman et al., 2008) were found for all four species. This includes data on development times, life span, and reproduction. Furthermore, many of these data types were reported for different constant experimental temperatures, providing sufficient information to calibrate a one-parameter Arrhenius temperature submodel, typically used in DEB models (Kooijman, 2010a).

However, data on growth over time, another essential data type for reliable DEB parametrization, were only found for *C. septempunctata* and *C. carnea* to a sufficient extent, complimented by the additional growth experiments performed for this study. The growth curve of the organism is not only a central feature that the model attempts to represent, the shape of the curve also allows important quantitative conclusions to be

drawn about the ratio of energy assimilated relative to the maintenance costs per unit of structure which is essential for a reliable parameter estimation (van der Meer, 2006; Kooijman et al., 2008). If no growth data are available, some information about these processes may also be inferred from respiration data. Whereas for *A. rhopalosiph* volumetric estimates based on microscopic images at known ages by Muratori et al. (2004) could be used as a substitute for actual growth-over-time data (leading to relatively high uncertainty; further discussed below), and respiration data were available from the literature, for *T. pyri* no such data were available. This lack of essential calibration data at the current state ultimately led to the exclusion of *T. pyri* from parametrization in this study. It is worth mentioning that the generation of reliable growth or respiration data of *T. pyri* is significantly more difficult (compared to, e.g., *C. septempunctata* and *C. carnea*) due to its particularly small size.

3.2. Additional growth experiments

Results from the additional growth experiments with *C. septempunctata* and *C. carnea* are represented in Table 2, Fig. 3, and Fig. 4. The raw data are available as CSV files in the supplementary material. The experiments yielded uniform growth curves with low variability between replicated test runs, which emphasizes the reliability of the data and makes them well suited for DEB calibration.

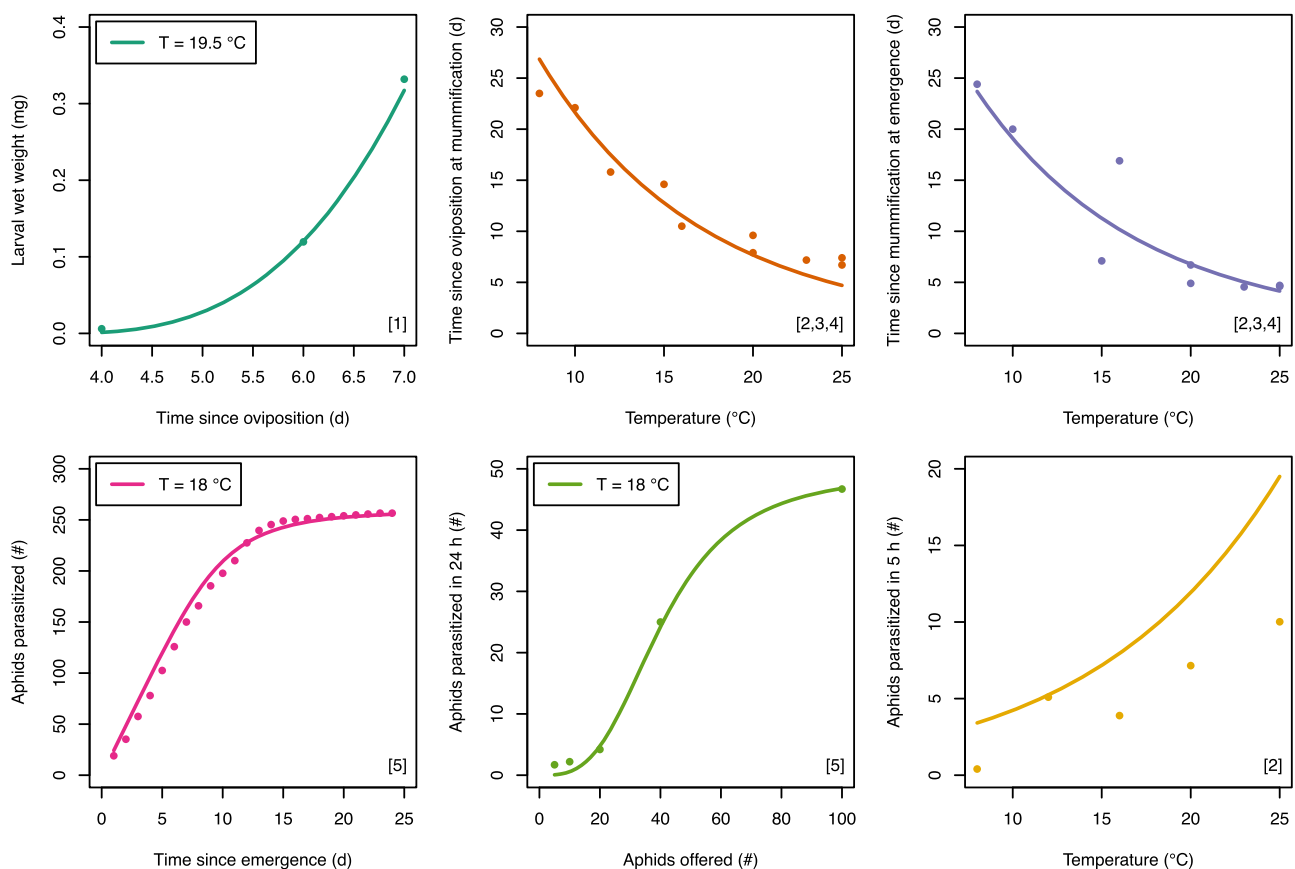
3.3. Evidence-based model selection

3.3.1. *T. pyri*

For *T. pyri* the *abj* model (Kooijman, 2010b) variant would have served as a starting point because currently all arachnids in the database are *abj* models. However, since this species was not considered further for a complete parametrization due to insufficient calibration data in the screened literature, no in-depth considerations were made about possibly necessary model refinements.

3.3.2. *A. rhopalosiph*

Since the OI of *A. rhopalosiph* was between 0.67 and 0.74 in an overview study by Jervis et al. (2001) analyzing data by Dransfield (1979), we made the simplifying model assumption that this species obtains all its reproductive energy during the larval stage. According to Bueno et al. (2012), a reduction of the parasitization rate over the female's lifespan (as can be seen for this species in Fig. 2; data by Shirota et al., 1983) is a characteristic commonly observed in pro-ovigenic species, which further supports this assumption. Therefore, both model variants *hex* and *hax* were compared, of which the *hax* model delivered the better fit to the data (See next section). As stated in the model description, imagines of the model variants *hex* and *hax* may or may not feed depending on the feeding behavior of the real organism. Since imagines of *A. rhopalosiph* feed on sugar sources, but do not grow and are not assumed to further invest energy in reproduction, it was



^[1]Muratori et al. (2004) (based on volumetric estimates), ^[2]Sigsgaard (2000), ^[3]Le Lann et al. (2011), ^[4]Fuentes-Contreras et al. (1996, 1998), ^[5]Shirota et al. (1983)

Fig. 2. Model fits to univariate life-history data for *A. rhopalosiph*. Experimental data are presented as dots whereas model simulations are presented as solid lines. The DEB model variant *hax* was parametrized to match the shown data as closely as possible. As an exception, the temperature-dependent parasitization rate data by Sigsgaard (2000) (bottom-right) were ignored during the parameter estimation in order not to distort the host-density-dependency curve by Shirota et al. (1983) (bottom center) to which they show a direct quantitative offset. Food availability in abundance ($f = 1$) was assumed for all simulations shown in this figure (Muratori et al., 2004; Sigsgaard, 2000; Le Lann et al., 2011; Fuentes-Contreras et al., 1996, 1998; Shirota et al., 1983). Note that additional zero-variate data were included in the parametrization that are not shown in this figure.

assumed that they feed at a rate that matches their maintenance costs, without any overhead for growth or reproduction.

3.3.3. *C. septempunctata*

Because in *C. septempunctata* the development of ovaries takes place mainly after the emergence of the female imago (e.g., Phoofole et al., 1995), this species' OI should be zero or very close to zero (extremely synovigenic). We therefore assumed that energetic investment in the reproduction buffer also takes place mainly after emergence. We see this assumption strengthened by the fact that adult ladybirds continue to feed predominantly on aphids, a protein-rich food that likely supports the continuous formation of eggs (Schmuck et al., 2000). Consequently, we modeled *C. septempunctata* using the newly developed model variant for synovigenic species.

3.3.4. *C. carnea*

In the case of *C. carnea*, a clear assignment between pro-ovigenic and synovigenic investment in reproduction was not immediately obvious based on literature information. Whereas initially, the hex and hax model variants were explored to model this species, all parametrization attempts led either to a greatly reduced lifetime fecundity (factor $\sim 1/500$) or unrealistically small eggs (factor $\sim 1/500$). This indicates that in the model not enough energy could be collected during the larval phase to achieve the fecundity numbers described in the literature (~ 700 – 1300 ; Syed et al., 2008; Amarasekare et al., 2013b) at realistic egg weights and associated costs per egg. This is a strong indication that a synovigenic model is better suited to describe the life cycle of *C. carnea*, as continuous feeding and investment in reproduction during the imago stage allows for larger numbers of eggs to be produced (Rosenheim, 2000). Therefore, we modeled the life cycle of *C. carnea* using the same new model variant for synovigenic species originally developed for *C. septempunctata*.

3.4. Model calibration outcome and considerations for further refinement

A list of the core DEB parameters calibrated for each of the three species is provided in Table 1. Additionally, some auxiliary parameters were needed to link model state variables to real life observables. A

complete list of all model parameters calibrated in this study, including auxiliary parameters and the additional model parameters used in the parasitization module for *Aphidius rhopalosiphii* can be found in the supplementary material or can be accessed from the model entries uploaded to the AmP portal (hyperlink: https://www.bio.vu.nl/thb/deb/deblab/add_my_pet/) (Koch et al., 2025a, 2025b; Albrecht et al., 2025).

Model fits to the univariate experimental data are presented in Figs. 2–4. Model fits to the most important zero-variate data are shown in Table 2. For reasons of conciseness, only the data that were considered most influential for calibrating the core parameters are listed there. A complete list of all zero-variate data included in the parametrization and associated model simulation results can be obtained for each respective species by accessing its entry in the AmP portal (hyperlink: https://www.bio.vu.nl/thb/deb/deblab/add_my_pet/) (Koch et al., 2025a; 2025b; Albrecht et al., 2025).

3.4.1. *A. rhopalosiphii*

Whereas after independent parametrization of both predefined model types hex and hax, most data could be reproduced well by both model variants, the pupal development times were considerably underestimated by the hex model (results shown in the supplementary material), leading to the exclusion of this model variant and the hax model being considered the definitive model variant for this species.

Overall, most data were simulated well by the hax model for *A. rhopalosiphii* as indicated by low relative errors and a good visual match to the univariate data. The growth curve in Fig. 2 in particular shows a strong visual and quantitative match ($RE = 0.04$) of observed versus simulated data. It should, however, be noted, that only three observations were available, which gives the model a relatively high degree of freedom when optimizing parameters to fit the curve. Furthermore, this data set should not be overinterpreted, since the sample size per data point was $n = 1$, and all data were subject to a relatively high degree of uncertainty as they were based on volumetric estimates from microscopic images.

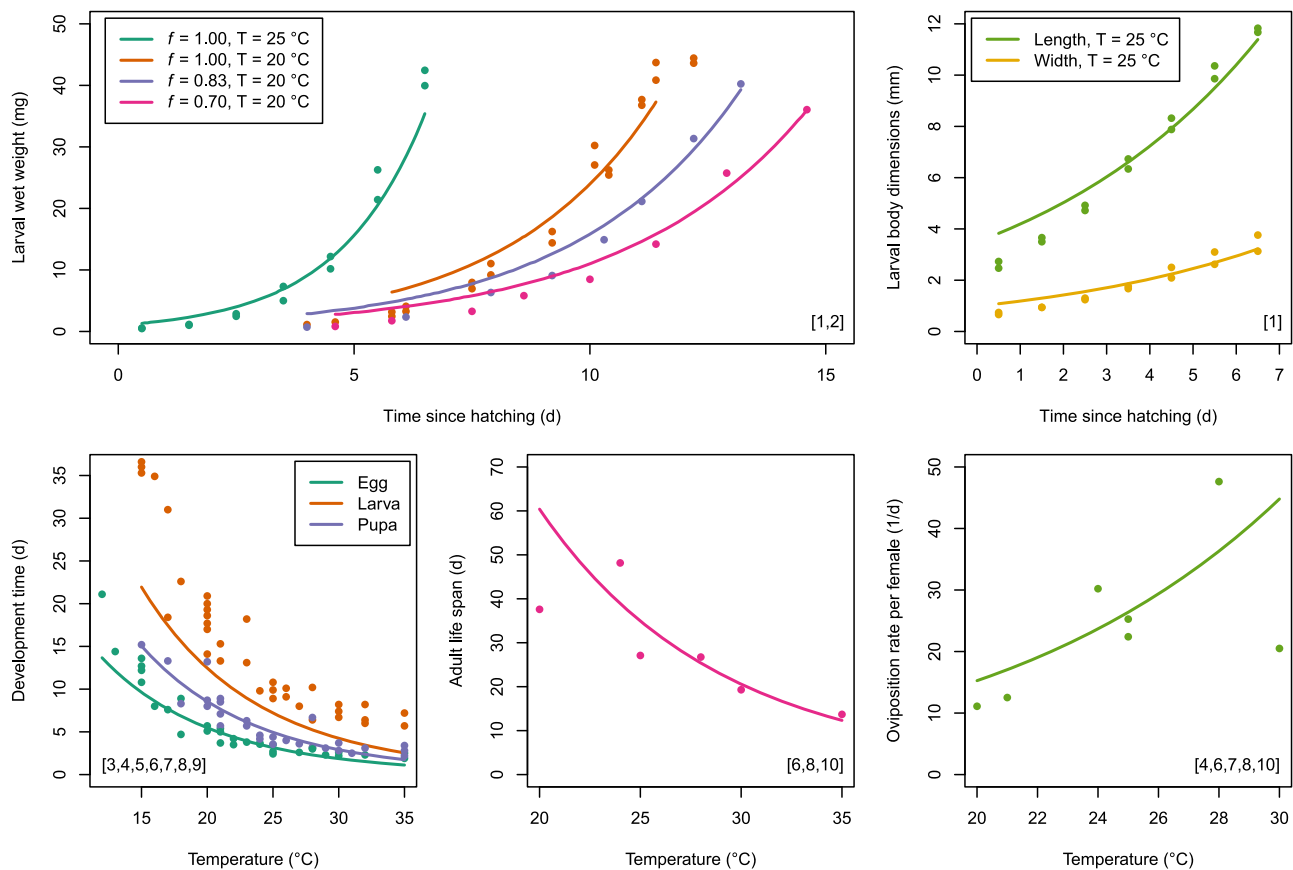
Already during the literature search it was found that there is a lack of growth data of the larval stage of *A. rhopalosiphii*. This is likely due to the fact that the larvae develop inside the host organism, and size and

Table 1

Core DEB parameters calibrated to the life-history data of *A. rhopalosiphii*, *C. septempunctata*, and *C. carnea* using either the model variant hax (*A. rhopalosiphii*) or the new DEB model variant for synovigenic holometabolous insects (*C. septempunctata* and *C. carnea*) introduced in this study. Note that direct quantitative comparisons of isolated parameters between the species are generally not conclusive, as some parameters are interdependent and some play slightly different roles in the different model variants.

Symbol	Unit	Definition	<i>A. rhopalosiphii</i>	<i>C. septempunctata</i>	<i>C. carnea</i>
$\{\dot{p}_{Am}\}$	J/d/cm ²	Surface-area-specific maximum assimilation rate	246	264	139
$\dot{v}$	cm/d	Energy conductance	1.55×10^{-2}	4.93×10^{-2}	2.53×10^{-2}
κ	–	Allocation fraction to soma	0.957	0.795	0.95
κ_X	–	Digestion efficiency of food to reserve	0.8 [a]	0.8 [a]	0.8 [a]
κ_R	–	Reproduction efficiency	0.95 [a]	0.95 [a]	0.95 [a]
κ_V	–	Total conversion efficiency from <i>E</i> to <i>V</i> (during growth), and back to <i>E</i> (during transformation at pupation)	0.7 [a]	0.7 [a]	0.8 [a]
$\dot{p}_M$	J/d/cm ³	Volume-specific somatic maintenance rate	748	47.8	17.7
$\dot{k}_J$	1/d	Maturity maintenance rate coefficient	0.002 [a]	0.002 [a]	0.002 [a]
$[E_G]$	J/cm ³	Volume-specific costs for growth	4.4×10^3 [a]	4.4×10^3 [a]	4.4×10^3 [a]
E_H^b	J	Maturity threshold for birth	7.59×10^{-7}	0.498	5.04×10^{-2}
E_H^j	J	Maturity threshold for pupation	NA	25.6	1.27
$[E_R^j]$	J/cm ³	Reproduction buffer density threshold for pupation	235	NA	NA
E_H^e	J	Maturity threshold for emergence	9.17×10^{-2}	34.4	2.28
E_H^p	J	Maturity threshold for puberty	2.00×10^{-5}	932	75.0
$\dot{h}_a$	1/d ²	Weibull aging acceleration	5.12×10^{-3}	3.64×10^{-5}	2.72×10^{-5}
s_G	–	Gompertz stress coefficient	1×10^{-4} [a]	1×10^{-4} [a]	1×10^{-4} [a]
T_A	K	Arrhenius temperature	8.60×10^3	9.57×10^3	1.24×10^4

[a] Some parameters were set to typical default values. This is common practice in DEB parametrization when it is assumed that the respective parameter cannot be reliably estimated based on the available data.



^[1]This study, ^[2]Blackman (1967), ^[3]Triltsch et al. (1996), ^[4]Sundby (1966), ^[5]Obrycki & Orr (1990), ^[6]Farooq et al. (2020), ^[7]Omkar & Srivastava (2003), ^[8]Singh & Singh (1993), ^[9]Kalushkov & Hodek (2004), ^[10]Xia et al. (1999)

Fig. 3. Model fits to univariate life-history data for *C. septempunctata*. Experimental data are presented as dots whereas model simulations are presented as solid lines. The new DEB model variant for synovigenic holometabolous insects was parametrized to match the shown data as closely as possible. Food availability in abundance ($f = 1$) was assumed for all simulations shown in this figure (Triltsch et al., 1996; Sundby, 1966; Obrycki and Orr, 1990; Farooq et al., 2020; Omkar and Srivastava, 2003; Singh and Singh, 1993; Kalushkov and Hodek, 2004; Xia et al., 1999) except when simulating growth curves at suboptimal feeding conditions as presented by Blackman (1967) (top left). Note that additional zero-variate data were included in the parametrization that are not shown in this figure.

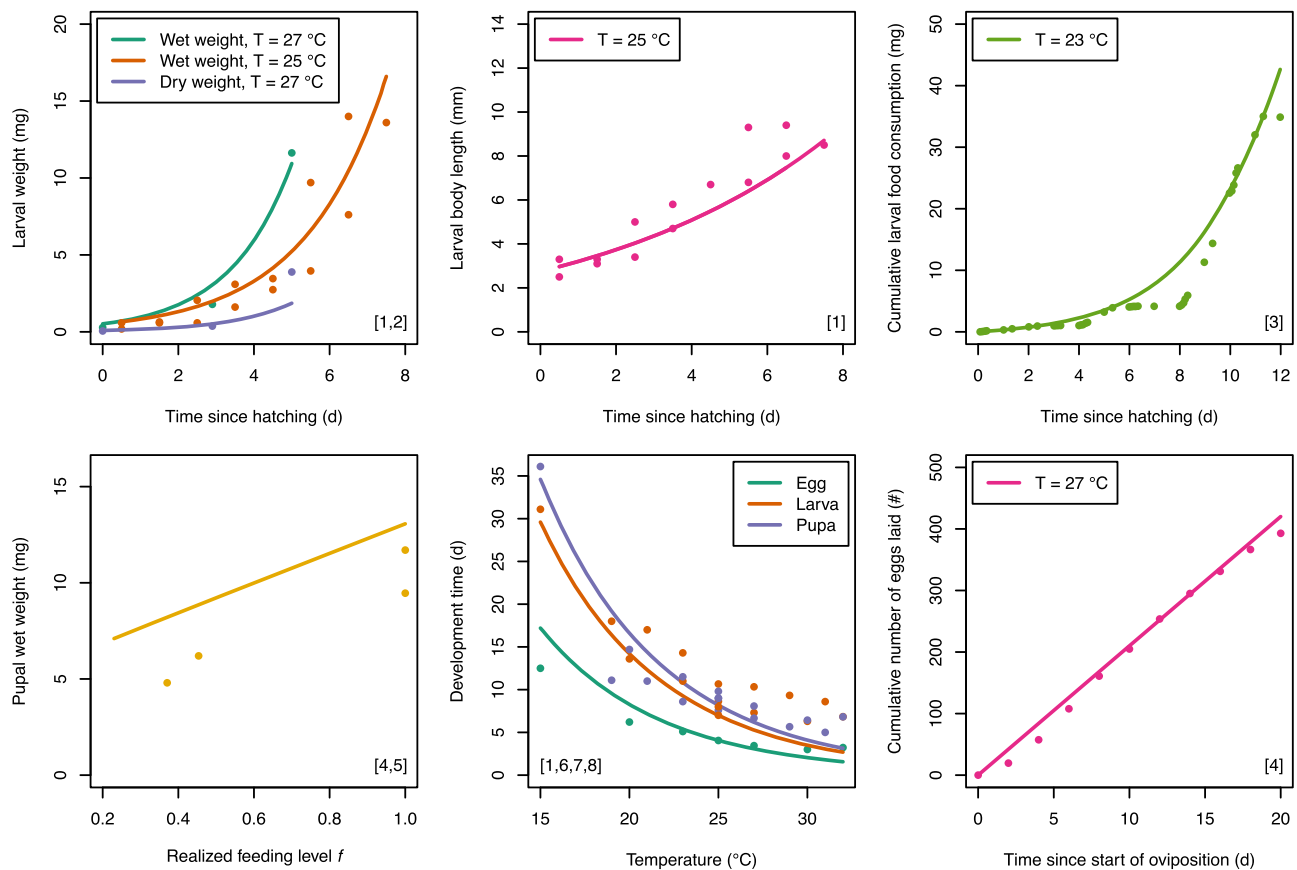
weight can only be determined by dissection of the hosts. This has the major disadvantage that sampling is always destructive (both the wasp larvae and the host are sacrificed in the process and are not available for future measurements), making growth data generation costly. For this parametrization, we used volumetric estimates based on microscopy images as a better-than-nothing solution. To enhance the model's parameterization with more reliable data, it would be beneficial to conduct a dedicated growth experiment that begins with a sufficiently large initial sample size of aphids shortly after parasitization, ensuring that enough replicates remain available for subsequent measurements, even after destructive sampling. Alternatively, further thought should be given to possible methods that allow non-invasive measurement, if this is technically possible in any way.

Data that were presented less well include the wet weight (over-predicted) and dry weight (under-predicted) of emerged females in Table 1. However, a closer look at the data reveals that the observed adult dry weight by Le Lann et al. (2012) was 0.82 times the observed adult wet weight by Le Lann et al. (2011), suggesting a water content of less than 20 % in fresh biomass, which appears unrealistic and hints at experimental variation between both studies. By contrast, the assumed dry weight per wet weight in the DEB model (based on DEBtool's default assumptions) is 0.17. Thus, the not-so-good fit can be attributed to a data inconsistency rather than a modelling issue.

Parasitization rates of *A. rhopalosiphii* in dependence of the time since adult emergence, and as a function of host abundance (2–5 days after

emergence) were generally met well by the model, and relative errors were low. Regarding the cumulative parasitization rate over time, a slight overestimation by the model until day 12 can be observed, which can be attributed to a slight overestimation of the parasitization on day 1, which is propagated over time due to the cumulative presentation of the data. Here the model assumes that the maximum parasitization rate is achieved from day one, whereas in reality there may be a minor start-up delay. However, we do not consider this difference substantial enough to further refine the parasitization module, which would further increase the number of parameters required.

Regarding the Holling Type III functional response of the parasitization rate as a function of host abundance, the observed parasitization rate at abundances of 5 and 10 aphids per setup was underestimated by the model. This is again because a rather simple function with only two parameters was used for the fit, and the low-abundance data do not perfectly match the Holling Type III curve inferred from the higher abundances. For model applications where low host abundances are important, the model should be refined if necessary to more accurately reflect the curve shape. However, in this case it would also be advisable to collect additional data, as the observed mismatch for the observations at low-abundance may simply be the result of experimental variation. Generally, we recommend that the density-dependent parasitization curve should be recalibrated (or potentially modified to a different curve shape) with new data when the modeled study design deviates from the design used by Shirota et al. (1983). This is because important



^[1]This study, ^[2]Zheng et al. (1993b), ^[3]Lawo & Romeis (2008), ^[4]Zheng et al. (1993a), ^[5]Jokar & Zarabi (2012), ^[6]Aghdam & Nemati (2020), ^[7]Amarasekare & Shearer (2013b), ^[8]Sharif et al. (2023)

Fig. 4. Model fits to univariate life-history data for *C. carnea*. Experimental data are presented as dots whereas model simulations are presented as solid lines. The new DEB model variant for synovigenic holometabolous insects was parametrized to match the shown data as closely as possible. Food availability in abundance ($f=1$) was assumed for all simulations shown in this figure (Lawo and Romeis, 2008; Zheng et al., 1993a; Jokar and Zarabi, 2012; Aghdam and Nemati, 2020; Amarasekare and Shearer, 2013b; Sharif et al., 2023) except when simulating food-consumption-dependent pupal wet weights as presented by Zheng et al. (1993b) (bottom left). Note that additional zero-variate data were included in the parametrization that are not shown in this figure.

experimental factors such as spatial confinement may have an effect on the curve shape. For example, Shiota et al. (1983) attributed the findings of Dransfield (1979), who had observed a Holling Type II response for the parasitization rate of *A. uzbekistanicus*, solely to a difference in cage size, arguing that it obscured the Holling Type III relationship by forcing the parasitoids to revisit the same locations within the cage more often.

The 5-h parasitization rate of *A. rhopalosiphii* (Fig. 2, bottom right) was set to be non-influential during calibration in order not to distort the host-density-dependency curve by Shiota et al. (1983) (bottom center) to which they show a direct quantitative offset. Therefore, a worse match to those data was expected. However, whereas the parasitization rates are generally overestimated by the model (again, as a result of in-between-study inconsistencies), the quantitative temperature dependency is represented relatively well by the model upon visual examination.

Overall, the parasitization module could adequately reproduce all the patterns that were taken into account during its development. However, it should be emphasized that, for simplicity, it does not yet account for some more complex feedback dynamics observed in Aphidiidae, such as those linking suboptimal adult nutrition to complex trade-offs between reproductive performance and longevity, as described by Tylianakis et al. (2004), and the above mentioned spatial dependencies which can affect the shape of the functional response curve. Depending on the model purpose, such mechanisms may still be

added in the future.

Since the parasitization module was designed as a stand-alone module that is not directly linked to the species-specific DEB parameters, but only requires the available energy in the reproduction buffer at the time of emergence and the energetic cost per egg as inputs, its assumptions had no significant influence on the calibration of the core DEB parameters. It can therefore be further refined if necessary without the need to reparametrize the core model. In principle, it can also be used modularly for other DEB models for parasitic wasp species that show comparable parasitization patterns.

3.4.2. *C. septempunctata* and *C. carnea*

Model fits to both zero-variate and univariate data were mostly acceptable in the models for *C. septempunctata* and *C. carnea* as indicated by low relative errors and a good visual match to the univariate data. Smaller deviations can be explained to a large extent by natural variation in the data. However, some systematic deviations were also identified, which are discussed hereinafter.

Notably, the initial size and weight of the organisms were overestimated in both models as seen in the zero-variate data (wet weight of an egg in *C. septempunctata* and wet weight at hatching in *C. carnea* in Table 2) and the univariate data of *C. septempunctata* (initial phase of the growth curves in Fig. 3). A likely explanation for the discrepancy is that the reproduction rates of adult females would have been overestimated if eggs were lighter, and thus energetically cheaper. The shortcoming in

Table 2

Model fit to zero-variate life-history data for all three newly parametrized models. Food availability in abundance ($f = 1$) was assumed for all simulations represented in this table. Note that information on temperature is only presented for data that are related to developmental times and rates, since other processes are temperature-independent in the DEB theory by default.

Data description	Temperature (°C)	Observed	Simulated	Mean relative error
<i>Aphidius rhopalosiph</i>				
Lifespan of an imago (d)	18	13.1 ^[1]	13.1	< 0.01
Wet weight of an egg (ng)	–	20.3 ^[2]	19.9	0.02
Wet weight of a mummy (mg)	–	0.42 ^[3]	0.688	0.64
Dry weight of a mummy (mg)	–	0.118 ^[2] 0.16 ^[3]	0.117	0.14
Dry weight at emergence (mg)	–	0.181 ^[2]	0.072	0.60
Wet weight at emergence (mg)	–	0.22 ^[2]	0.422	0.92
Mean basal metabolic rate ($\mu\text{L CO}_2/\text{h}$)	20	0.41 ^[2]	0.408	< 0.01
<i>Coccinella septempunctata</i>				
Pre-oviposition period (d)	24	3.26 ^[4]	7.38	1.26
Pre-oviposition period (d)	25	8.2 ^[5]	6.63	0.19
Pre-oviposition period (d)	28	6.66 ^[6]	4.81	0.28
Wet weight of an egg (μg)	–	187 ^[7]	1213	5.49
Pupal wet weight (mg)	–	15.05 ^[5] 42.0 ^[8]	52.0	1.35
Wet weight at emergence (mg)	–	37.7 ^[9] 38.8 ^[10] 40.6 ^[5] 37.5 ^[8] 65.9 ^[8]	33.4	0.13
Wet weight > 7 d after emergence (mg)	–		69.2	0.05
Dry-weight-specific assimilation rate at first instar (J/mg/d)	27	25.0 ^[11]	18.6	0.26
Dry-weight-specific assimilation rate at second instar (J/mg/d)	27	29.8 ^[11]	18.6	0.38
Dry-weight-specific assimilation rate at third instar (J/mg/d)	27	37.2 ^[11]	18.6	0.50
Dry-weight-specific assimilation rate at fourth instar (J/mg/d)	27	37.3 ^[11]	18.6	0.50
Dry-weight-specific assimilation rate of imago (J/mg/d)	27	49.2 ^[11]	35.0	0.29
<i>Chrysoperla carnea</i>				
Pre-oviposition period (d)	23	4.2 ^[12]	4.21	< 0.01
Lifespan of an imago (d)	23	51.7 ^[12] 48.3 ^[13]	49.9	0.03
Mean reproduction rate (#/d)	23	27.7 ^[12] 14.4 ^[13]	12.0	0.37
Mean reproduction rate (#/d)	25	15.67 ^[14] 8.67 ^[15]	15.9	0.43
Mean reproduction rate (#/d)	32	27.37 ^[16]	41.4	0.51
Wet weight at hatching (mg)	–	0.1 ^[17]	0.521	4.21
Wet weight of a pupa (mg)	–	8.3 ^[18]	13.1	0.57
Wet weight at emergence (mg)	–	10.3 ^[8]	10.4	0.01

Table 2 (continued)

Data description	Temperature (°C)	Observed	Simulated	Mean relative error
Wet weight > 4 d after emergence (mg)	–	29.7 ^[8]	23.2	0.22
Dry weight at emergence (mg)	–	3.11 ^[19]	1.76	0.43

^[1] Shiota et al. (1983).

^[2] Le Lann et al. (2012).

^[3] Legrand et al. (2004).

^[4] Farooq et al. (2020).

^[5] Omkar & Srivastava (2003).

^[6] Singh & Singh (1993).

^[7] Chatha & Naz (2020).

^[8] This study.

^[9] Obrycki & Orr (1990).

^[10] Kalushkov & Hodek (2004).

^[11] Singhal (1976).

^[12] Amarasekare & Shearer (2013b).

^[13] Amarasekare & Shearer (2013a).

^[14] Jokar & Zarabi (2012).

^[15] Sharif et al. (2023).

^[16] Chakraborty & Korat (2010).

^[17] Baumgartner et al. (1981).

^[18] Syed et al. (2008).

^[19] Lawo & Romeis (2008).

Relative error = absolute difference (observed data minus simulated data) divided by observed data; e.g., 0 = perfect fit, 0.01 = 1 % deviation, 1 = 100 % deviation.

the model therefore seems to be that adult animals have an unrealistically high investment in reproduction under the standard assumption that their parameters for the specific assimilation rate and maintenance costs remain unchanged from the larval stage. To resolve this, either a reduced assimilation rate, or higher maintenance costs for the imago could be assumed, implemented via an additional free model parameter. Since the somatic maintenance costs are supposed to represent all basic energetic costs of living of the organism, a plausible explanation would be that an increase in these costs is attributed to the costs of flying, not present in the larvae. However, since this is only a first intuitive explanation, and our aim was to keep the number of free model parameters as low as possible, we have refrained from implementing this model refinement for the time being. Potentially, respiration tests could be carried out with different life stages, which would help to better quantify the maintenance expenses, and help to make a well informed modelling decision.

Furthermore, data on larval development times were mostly underestimated by the model. This seems to be due to the fact that the pupal weights would be significantly overestimated if the larval phase and thus the steep larval growth were to last longer. It has been reported for holometabolous insects that in reality they undergo significant behavioral changes shortly before pupation. These include a reduction or complete stop of feeding, which goes hand in hand with a reduction in growth (Browne, 1995). Reportedly, such feeding pauses also occur to a small extent before each larval transition (e.g., reflected in the cumulative feeding curve of *C. carnea* in Fig. 4; data by Lawo and Romeis, 2008). Whereas in theory, each feeding pause could be modeled explicitly, we consider it acceptable to assume averaged, uninterrupted feeding throughout the larval growth phase for the sake of model simplicity. However, in order to be able to estimate larval development times more realistically, it would be conceivable to at least include the final growth stop before pupation in a potential model refinement. While such a model refinement could take various forms, it seems simplest to introduce an additional maturity threshold that triggers an additional functional life stage for the late larva. In this stage, the larva

would not feed (or feed at reduced rate, requiring yet an additional model parameter), and the growth might decline either *naturally* as a result of the reduced food uptake, or be forced to an abrupt halt (using the same maturity threshold), whichever seems most realistic based on real observations.

Apart from these minor model inaccuracies, we consider the new model structure to be suitable for representing the complex life cycle of both insect species in a DEB model. Furthermore, we find it notable that the newly created model structure, which initially was designed specifically to fit the life cycle of *C. septempunctata*, also provided good fits for *C. carnea* without the need for further model modifications. This is an indication that the structure could also be applicable to other insect species. If a broader applicability were confirmed, the model could potentially be established as a go-to template for synovigenic insect species in general, filling an important gap in the toolbox of predefined DEB variants. However, it is also noticeable that the model had the same predictive weaknesses for both species, which could likely be eliminated by further model refinements as discussed above.

In this study, however, we have placed particular emphasis on ensuring that the model structure is not more complex than necessary and that the number of model parameters is kept as low as possible. Nevertheless, if the minor model discrepancies identified in this study also apply to other species, it should be carefully considered whether a slightly higher model complexity in favor of more realistic model simulations would be useful when attempting to establish a new generic model variant. We therefore consider it a logical next step to further test the model structure using new case examples to determine whether it is suitable as a generic model variant or whether a higher model complexity is required. Overall, care should be taken to ensure that the model is not tailored too specifically to individual species, as otherwise the general applicability for a larger species pool could suffer.

Notably, the submission of the new model entries for *C. septempunctata* and *C. carnea* to the AmP portal (hyperlink: https://www.bio.vu.nl/thb/deb/deblab/add_my_pet/) has already led to new fruitful exchanges with the curators of the database on how the model structure can be further optimized, e.g., by eliminating complex constraints of the parameter space through some additional model assumptions, making the parameter estimation process more robust against unrealistic parameter combinations. As this is an ongoing discussion, and further adjustments may well be made in the future, we recommend that you always consult the latest version of the database entries in the AmP portal (hyperlink: https://www.bio.vu.nl/thb/deb/deblab/add_my_pet/), as well as the latest version of the DEB documentation (Kooijman, 2010b).

4. Conclusions

In this study, we were able to develop DEB models for three out of the four selected NTA species, which could be successfully calibrated with life-history data mostly obtained from the open literature. The literature searches have shown, however, that especially for species that are difficult to measure during the larval stage, either due to their small size (*T. pyri*) or because they grow parasitically within the host animal (*A. rhopalosiphi*), growth data can be difficult to find, which can make a robust DEB parametrization difficult or even impossible.

Whereas the life cycle of *A. rhopalosiphi* could be reproduced well with the hax model, one of the DEB model variants predefined in the DEB literature, none of the predefined DEB model variants could adequately represent the life cycles of *C. septempunctata* and *C. carnea*, both of which we considered to be extremely synovigenic holometabolous insects. However, we were able to demonstrate that also the life cycles of synovigenic holometabolous insects can be reproduced using the DEB theory, by recombining elements from the existing models, resulting in a novel model structure. Further testing will show if this new model variant is also applicable to other species. Regardless, we see the model structure presented here as an important starting point for DEB

modeling of synovigenic insects, whether in a regulatory context or elsewhere.

Although the models presented in this study, like all DEB models, can always be further refined, we consider their overall accuracy in simulating the data to be convincing. Furthermore, we consider them ideally suited for DEB-TKTD applications in their current form due to the carefully chosen balance between model accuracy and model complexity.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used the DeepL translation tool in order to improve sentence structure. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRediT authorship contribution statement

Josef Koch: Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Natalie Albrecht:** Writing – review & editing, Software, Methodology, Investigation, Formal analysis, Data curation. **Jana Gerhard:** Writing – review & editing, Software, Methodology, Investigation, Formal analysis, Data curation. **Kim Rakel:** Writing – review & editing, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Heike Fremdt:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Data curation, Conceptualization. **Daniela Jans:** Writing – review & editing, Resources, Investigation, Data curation. **André Gergs:** Writing – review & editing, Methodology, Conceptualization. **Thomas G. Preuss:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Josef Koch reports financial support, article publishing charges, and travel were provided by Bayer AG. Natalie Albrecht, Jana Gerhard, and Kim Rakel report financial support was provided by Bayer AG. Heike Fremdt, Daniela Jans, André Gergs, and Thomas G. Preuss report being employed by Bayer AG. Bayer AG produces and sells agrochemicals. Bayer AG applies DEB models to assess the effects of chemicals on target and non-target species and has an interest in getting these models accepted during regulatory submissions. Given his role as Guest Editor, André Gergs had no involvement in the peer-review of this article and has no access to information regarding its peer-review. Full responsibility for the editorial process for this article was delegated to Hsiao-Hsuan (Rose) Wang.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ecolmodel.2025.111440](https://doi.org/10.1016/j.ecolmodel.2025.111440).

Data availability

All experimental data collected in this study are provided as CSV files in the supplementary material. All code and data used for model calibration can be accessed via the Add-my-Pet portal.

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